

What are posterior predictive checks and what makes them useful?

Asked 10 years, 4 months ago Modified 7 years, 7 months ago Viewed 29k times

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I understand what the [posterior predictive distribution](#) is, and I have been reading about [posterior predictive checks](#), although it isn't clear to me what it does yet.

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1. What exactly is the posterior predictive check?

2. Why do some authors say that running posterior predictive checks is "using the data twice" and should not be abused ? (or even that it is not Bayesian)? (e.g. see [this](#) or [this](#))

3. What is this check exactly useful for? Can it really be used for model selection? (e.g. does it factor in both, fitness and model complexity?)

bayesian model-selection posterior

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edited Jan 30, 2015 at 2:43

Jeromy Anglim

45.8k ● 24 @ 156 ● 261

asked Sep 11, 2014 at 18:37

Amelio Vazquez-Reina

19.7k ● 27 @ 81 ● 120

1 Answer

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Posterior predictive checks are, in simple words, "simulating replicated data under the fitted model and then comparing these to the observed data" ([Gelman and Hill, 2007, p. 158](#)). So, you use posterior predictive to "look for systematic discrepancies between real and simulated data" ([Gelman et al, 2004, p. 169](#)).

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The argument about "using the data twice" is that you use your data for estimating the model and then, for checking if the model fits the data, while generally it is a bad idea and it would be better to [validate your model on external data](#), that was not used for estimation.

Posterior predictive checks are helpful in assessing if your model gives you "valid" predictions about the reality - do they fit the observed data or not. It is a helpful phase of model building and checking. It does not give you a definite answer on if your model is "ok" or if it is "better" than other model, however, it can help you to check if your model makes sens.

This is nicely described in **LaplaceDemon** vignette [Bayesian Inference](#):

Comparing the predictive distribution y^{PP} to the observed data y is generally termed a "posterior predictive check". This type of check includes the uncertainty associated with the estimated parameters of the model, unlike frequentist statistics.

Posterior predictive checks (via the predictive distribution) involve a double-use of the data, which violates the likelihood principle. However, arguments have been made in favor of posterior predictive checks, provided that usage is limited to measures of discrepancy to study model adequacy, not for model comparison and inference (Meng 1994).

Gelman recommends at the most basic level to compare y^{PP} to y , looking for any systematic differences, which could indicate potential failings of the model (Gelman et al. 2004, p. 159). It is often first recommended to compare graphical plots, such as the distribution of y and y^{PP} .

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edited May 30, 2017 at 14:20

answered Nov 26, 2014 at 14:21

Tim

141k ● 26 @ 271 ● 513

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regarding your aside that PPC's might not be Bayesian, note that Gelman discusses a lot what exactly a Bayesian/Bayesian data analysis is [Gelman and Shalizi](#)

– N Brouwer Dec 20, 2016 at 15:10
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Gelman & Shalizi discuss what Bayesian data analysis is according to *their* point of view. There are many different points of view out there, all well motivated – from [Jeffreys](#), to Savage, from de Finetti to Gelfman, [Scott & Krauss](#) & [Hailperin](#), not to mention views differing even more, like [Demoster-Shafer's](#) – pg10m Sep 30, 2017 at 21:42